Lecture Scribe System Reliability

L23_Amazone Sale_2026_...
PDF

You are required to generate a lecture scribe intended strictly as exam-oriented reference material. Use ONLY the provided context (lecture slides / lecture PDF). Do NOT introduce any content, examples, explanations, intuition, or results that do not explicitly appear in the provided context. While writing the lecture scribe, reason step by step through the material exactly as it is presented in the lecture. Include explicit reasoning in the scribe wherever the lecture contains: - definitions and notation - assumptions and conditions - theorem or result statements - proofs or proof sketches - derivations - worked examples All reasoning must be faithful to the provided context and must follow the logical order used in the lecture. Make the logical dependencies between concepts explicit so that a student can reconstruct the lecture while revising for an exam. Maintain a formal, academic tone suitable for open-notes exam preparation. Organize the scribe clearly with appropriate section headings. Do NOT: - invent new examples or explanations - simplify or reinterpret arguments - add intuition not discussed in the lecture - go beyond the scope of the provided context Ensure a clear and logical progression of ideas to facilitate student understanding during revision. This should serve as concise and accurate exam reference material. Also cover each and every topic do NOT skip any single topic or examples. Get notes for handwritten content too.

Lecture Scribe: Tail Bounds and System Reliability (Case Study: Amazon Black Friday Sale 2026)

Course: CSE400 – Fundamentals of Probability in Computing

Lecture 23

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1. Introduction: System Reliability under Uncertainty

1.1 Context: Amazon Black Friday Sale 2026

From the lecture (page 2):

- Expected sale timeline:
  - Early deals: November 20–21, 2026
  - Main sale: November 25–27, 2026
  - Cyber Monday: November 30, 2026
- System characteristics:
  - High demand across categories (electronics, devices, apparel, home appliances)
  - Additional load due to:
    - Prime early access
    - Bank discounts and cashback offers

Logical implication:
The system must handle **extreme traffic spikes**, not just average load.

1.2 Connection to Computing and Uncertainty

From the diagram (page 3):

- Server system with multiple servers handling jobs.
- Highlighted condition:
  - A server has ≥ **k jobs**
- Arrival model:
  - **Poisson arrivals (in datacenter)**
  - Example:
    - ≥ k arrivals within 1 hour

Conclusion from lecture structure:
System load is modeled probabilistically, focusing on **rare high-load events**.

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2. Problem Formulation: Tail Probability and Failure

From page 4:

- Key quantity:

$$P(X \geq k)$$

- Interpretation:
  - Probability that system load exceeds threshold k
  - This corresponds to:
    - **System failure event**
    - **Overload threshold breach**

2.1 Key Insight

From page 5:

- "The mean is safe. The tail is catastrophic."
- Simulation:
  - Poisson traffic with parameter:

$$\lambda = 50$$

  - Crash threshold:

$$k = 60$$

- Observations:
  - Even when mean ≈ 50, spikes exceed 60
  - Empirical crash rate: **8.7% (87 crashes / 1000 minutes)**

Conclusion:

- Mean behavior does not capture failure risk.
- Need to compute:

$$P(X \geq 60)$$

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3. Motivation for Bounds

From page 6:

3.1 Core Insight

- Exact probability may not always be computable.
- Use **upper bounds** instead.

3.2 Hierarchy of Bounds

Level	Method	Input Required
Level 1	Markov	Mean (μ)
Level 2	Chebyshev	Variance (σ^2)
Level 3	Chernoff	Full distribution (MGF)

Conclusion:

- More information ⇒ tighter bounds.

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4. Markov's Inequality

4.1 Statement (from page 7)

$$P(X \geq a) \leq \frac{\mathbb{E}[X]}{a}$$

4.2 Requirements

- Only requires:

$$\mathbb{E}[X] = \mu$$

4.3 Behavior

- Decay:

$$\propto \frac{1}{k}$$

  (Linear decay)

4.4 Observation from Graph

- Markov bound significantly overestimates:
  - Example (page 7):
    - Exact: 0.004
    - Markov: 0.714

Conclusion:

- Very loose bound
- "Wastes space" due to overestimation

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5. Chebyshev's Inequality

5.1 Statement (from page 8)

$$P(|X-\mu| \geq a) \leq \frac{\sigma^2}{a^2}$$

5.2 Requirements

- Mean μ
- Variance σ^2

5.3 Behavior

- Decay:

$$\propto \frac{1}{k^2}$$

  (Polynomial decay)

5.4 Observation from Graph

- Example (page 8):
  - Exact: 0.004
  - Chebyshev: 0.125

Conclusion:

- Tighter than Markov
- Still overestimates

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6. Chernoff Bound

6.1 Statement (from page 9)

$$P(X \geq a) \leq \min_{t>0} e^{-ta} M_X(t)$$

where:

$$M_X(t) = \mathbb{E}[e^{tX}]$$

6.2 Requirements

- Full distribution via:
  - Moment Generating Function (MGF)

6.3 Behavior

- Decay:

$$\propto e^{-k}$$

  (Exponential decay)

6.4 Observation from Graph

- Example (page 9):
  - Exact: 0.092
  - Chernoff: 0.391 (tightest among bounds)

Conclusion:

- Tightest bound among the three
- Closely follows actual tail probability

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7. Comparison of Decay Rates

From page 10:

Method	Decay Type	Expression
Markov	Linear	$1/k$
Chebyshev	Polynomial	$1/k^2$
Chernoff	Exponential	e^{-k}

Key statement:

- Chernoff decays fastest
- Markov remains highly pessimistic

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8. Case Study: SLA Breach

From page 11:

8.1 Parameters

- Distribution:

$$X \sim \text{Poisson}(\lambda = 500)$$

- SLA threshold:

$$k = 600$$

8.2 Objective

- Compute:

$$P(X > 600)$$

- Use bounds to:
  - Size backup capacity

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9. Cost of Loose Bounds

From page 12:

Method	Estimated Failure Probability
Exact	~1.4%
Chernoff	~8.8%
Chebyshev	~20%
Markov	~90.9%

9.1 Interpretation

- Markov:
  - Predicts extremely high failure probability
  - Leads to over-provisioning
- Chernoff:
  - Closest to actual

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10. Financial Interpretation

From pages 13–14:

10.1 Waste Multiplier

$$\text{Waste Multiplier} = \frac{\text{Bound Estimate}}{\text{Exact Probability}}$$

10.2 Insight

- Overestimation ⇒ extra servers purchased
- "Bad math burns budget"

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11. Effect of Distribution Type

From page 15:

- Chernoff works well for:
  - Normal
  - Poisson
  - Binomial
- Limitation:
  - For **Gamma (bursty, heavy-tailed)**:
    - Chernoff can overestimate

Conclusion:

- Choice of bound depends on distribution

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12. Engineer's Decision Matrix

From page 16:

Method	Requires	Decay	Tightness	Use Case
Markov	Mean	Linear	Loose (10–1000x)	Only mean known
Chebyshev	Mean + Variance	Polynomial	Moderate (2–10x)	Variance known
Chernoff	Full MGF	Exponential	Tight (<2x)	Precise SLA modeling

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13. Class Participation: Laplace Distribution

From page 17:

13.1 Given

$$X \sim \text{Laplace}(\mu, b)$$

- Properties:
  - Sharp central peak
  - Heavy tails
  - Standard deviation:

$$\sigma = b\sqrt{2}$$

13.2 Objective

- Derive Chernoff bound
- Threshold:

$$a = \mu + 3\sigma$$

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14. Derivation Blueprint (Step-by-Step)

From page 18:

Step 1: Compute MGF

$$M_X(t) = \mathbb{E}[e^{tX}]$$

Step 2: Apply Chernoff Inequality

$$P(X \geq a) \leq e^{-ta} M_X(t)$$

Step 3: Optimize

- Differentiate w.r.t. t
- Set derivative = 0
- Solve for optimal t

Step 4: Substitute

- Plug optimal t into inequality
- Obtain tightest bound

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15. Final Logical Flow of Lecture

1. Real-world system (Amazon sale) → uncertainty
2. Model traffic using probability (Poisson)
3. Define failure via tail probability
4. Introduce bounds when exact probability unavailable

5. Compare:
    - Markov (mean only)
    - Chebyshev (variance)
    - Chernoff (MGF)
6. Evaluate impact on:
    - System reliability
    - Financial cost
7. Apply framework to new distribution (Laplace)

End of Lecture Scribe